

Chapter 1 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Which of the following metric relationships is incorrect?

- A) 1 microliter = 10^{-6} liter
- B) 10^3 grams = 1 kilogram
- C) 10^6 millimeters = 1 meter
- D) 1 gram = 10^2 centigrams
- E) 10 decimeters = 1 meter

2. Which of the following is not an exact number?

- A) the number of seconds in a year
- B) the number of millimeters in a kilometer
- C) the number of passengers in an airplane
- D) the volume of a cup of coffee
- E) the number of pages in a textbook

3. An air sample consists of oxygen and nitrogen gas as major components. It also contains carbon dioxide and traces of some rare gases. All these gases are evenly distributed throughout the sample of air.

Which term or terms could be used to describe this sample of air?

i) mixture; **ii)** heterogeneous mixture; **iii)** homogenous mixture; **iv)** solution; **v)** pure substance; **vi)** compound; **vii)** element

- A) i and ii
- B) i and iii
- C) i and iv
- D) i
- E) i, iii and iv

4. Which of the following is the least probable concerning five measurements taken in the lab?

- A) The measurements are accurate and precise.
- B) The measurements are accurate but not precise.
- C) The measurements are precise but not accurate.
- D) The measurements are neither accurate nor precise.
- E) All of these are equally probable.

5. In the spring of 2008, petrol cost £1.049 per litre in London. On the same day, the exchange rate was \$1 = £0.503. What was the price of London petrol in dollars (\$) per gallon? (1 gal = 3.7854 L)

- A) \$7.89 / gal
- B) \$3.97 / gal
- C) \$2.00 / gal
- D) \$1.82 / gal
- E) \$7.53 / gal

6. Which unit of measurements is not SI unit?

- A) m
- B) s
- C) mol
- D) g
- E) K

7. Which one of the following is an intensive property?

- A) mass
- B) temperature
- C) length
- D) volume
- E) amount

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8. Which of the following statements concerning the common temperature scales is/are true?

- i) Fahrenheit (°F) is an absolute temperature scale.
 - ii) The normal boiling point of water (100 °C) is equal to 273 K.
 - iii) The difference between the boiling point and freezing point of a substance is the same for the Celsius (°C) and the Kelvin (K) scales.
- A) i only
B) ii only
C) iii only
D) ii and iii
E) i, ii, and iii

9. All the following are examples of chemical changes except _____.

- A) evaporation
B) photosynthesis
C) fermentation
D) aging
E) respiration

10. For which of the following can the composition vary?

- A) pure substance
B) element
C) homogeneous mixture
D) heterogeneous mixture
E) both homogeneous and heterogeneous mixtures

Part B Short questions (Write legibly and show work for all steps in the following problems. For mathematical relationships, show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures.)

11. (2 points) Carry out the following operation and express the answer with the appropriate number of significant figures.

$$12.00000 \times 0.9893 + 13.00335 \times 0.0107$$

12. (5 points) Read the following description of the element zinc and indicate which are physical properties and which are chemical properties.

Zinc melts at 420 °C. When zinc granules are added to dilute sulfuric acid, hydrogen is given off and the metal dissolves. Zinc has a hardness on the Mohs scale of 2.5 and a density of 7.13 g/cm³ at 25 °C. It reacts slowly with oxygen gas at elevated temperatures to form zinc oxide, ZnO.

13. (3 points) Classify each of the following as an element, a compound, or a heterogeneous mixture: **(a)** Aspirin is composed of 60.0% carbon, 4.5% hydrogen, and 35.5% oxygen by mass; **(b)** rocks; **(c)** iodine crystals.

Chapter 1 Homework Answer Sheet

Name:

Student ID:

Instructor:

Score:

Part A Multiple choice (10 points)

1-5_____

6-10_____

Part B Short questions (10 points)